



DESIGNER'S REFERENCE HANDBOOK



AGC-4



DEIF A/S · Frisenborgvej 33 · DK-7800 Skive
Tel.: +45 9614 9614 · Fax: +45 9614 9615
info@deif.com · www.deif.com

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1. Introduction

1.1 About the Designer's Reference Handbook

1.1.1 General purpose

This **Designer's Reference Handbook** provides overall information about the controller applications and functions. This document also provides the information needed to configure the application and its parameters.

See the **Installation instructions** for installation information. See the **Operator's manual** for information on how to operate the controller.



CAUTION

Read this document before starting to work with the controller and the equipment to be controlled. Failure to do this could result in human injury or damage to the equipment.

1.1.2 Intended users

This Designer's Reference Handbook is mainly intended for the panel builder designer. On the basis of this document and the Installation instructions, the panel builder designer will give the electrician the information he needs to install the controller, for example, detailed electrical drawings.

1.1.3 Options

This **Designer's Reference Handbook** describes the standard AGC-4 controller.

The controller functions can be increased by a variety of flexible hardware and software options. The options include generator, busbar and mains protections; voltage/VAr/PF control; various outputs; power management; serial communication; additional operator display; and more. The **Data Sheet** includes a complete list of options.

The options are described in detail in **Description of Options** documents.



INFO

You can find all the AGC-4 documentation here: www.deif.com/products/ago-4#documentation

1.1.4 Parameter list

The Designer's Reference Handbook refers to parameters. For further information, see the **Parameter list**.

1.1.5 Glossary

Term	Abbreviation	Explanation
Additional Operator Panel	AOP	
AGC-4	AGC-4	A genset, BTB, or mains controller.
Automatic Load Controller	ALC-4	
Automatic Mains Failure	AMF	If there is a mains failure, the AGC automatically uses a genset to supply the load.
Automatic Sustainable Controller	ASC-4	
Automatic transfer switch	ATS	
Automatic voltage regulator	AVR	

Term	Abbreviation	Explanation
Busbar	BB	
Bus tie breaker	BTB	
Current transformer	CT	
Display unit	DU-2	
Genset	G	
Genset breaker	GB	
Governor	GOV	
Load take over	LTO	
Mains breaker	MB	
Mains power export	MPE	
Menu	(XXXX)	A group of parameters.
M-Logic		The PLC-type tool accessible from the utility software.
Multi-line-2	ML-2	A DEIF platform, which includes the AGC-4.
Nominal power	P nom	
Nominal reactive power	Q nom	
Nominal voltage	U nom	
Parameter	(XXXX)	A configurable setting (sometimes also called <i>Channel</i> in the PC utility software).
PC utility software	USW	
Resistance measurement input	RMI	
Software	SW	
Touch display unit	TDU 107	
Voltage transformer	VT	

1.2 Warnings, legal information and safety

1.2.1 Warnings and notes

Throughout this document, a number of warnings and notes with helpful user information will be presented. To ensure that these are noticed, they will be highlighted as follows in order to separate them from the general text.

Warnings



DANGER!

This highlights dangerous situations. If the guidelines are not followed, these situations could result in death, serious personal injury, and equipment damage or destruction.



CAUTION

This highlights potentially dangerous situations. If the guidelines are not followed, these situations could result in personal injury or damaged equipment.

Notes



INFO

Notes provide general information, which will be helpful for the reader to bear in mind.

1.2.2 Factory settings

The controller is delivered pre-programmed from the factory with a set of default settings. These settings are based on typical values and may not be correct for your system. You must therefore check all parameters before using the controller.

1.2.3 Automatic and remote-controlled starts

The power management system automatically starts gensets when more power is needed. It can be difficult for an inexperienced operator to predict which gensets will start. In addition, gensets can be started remotely (for example, by using an Ethernet connection, or a digital input). To avoid personal injury, the genset design, the layout, and maintenance procedures must take this into account.

1.2.4 Safety during installation and operation

Installing and operating the equipment may require work with dangerous currents and voltages. The installation must only be carried out by authorised personnel who understand the risks involved in working with electrical equipment.



DANGER!

Hazardous live currents and voltages. Do not touch any terminals, especially the AC measurement inputs and the relay terminals. Touching the terminals could lead to injury or death.

1.2.5 Electrostatic discharge awareness

Sufficient care must be taken to protect the terminal against static discharges during the installation. Once the unit is installed and connected, these precautions are no longer necessary.

1.3 Legal information and disclaimer

DEIF takes no responsibility for installation or operation of the generator set or switchgear. If there is any doubt about how to install or operate the engine/generator or switchgear controlled by the Multi-line 2 unit, the company responsible for the installation or the operation of the equipment must be contacted.



INFO

The Multi-line 2 unit is not to be opened by unauthorised personnel. If opened anyway, the warranty will be lost.

Disclaimer

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